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NATIONAL DEVELOPMENT COMPLEX
Deputy Directorate General of Quality Control
PHYSICAL LAB, Site-3A
TEST REPORT

Test Report No: TR (046)/PL/2022

Job Card No: JC (073)/PL/2022

Date: 13-05-2022

INDENTER'S DATA			
Name:	Muhammad Imran	Designation:	Research Associate
Group & Site:	Preform / 3T	Directorate/ Department:	3DC-P
Organization:	NDC	Contact No:	051-9314438 Ext. 209

GENERAL DATA			
Sample Name:	3D Carbon Composite	ID No:	RD-19C/Cp/01~03
Lot No :	3D/CCC/Jun21/L-09	Lot Size:	----
Shape:	Chips	Chemical Composition:	Carbon
Project:	AbWS/S1/S2	Quantity:	03
Sample Receiving Date:	11-05-2022	Test Date:	12, 13-05-2022

TEST DATA			
Test Method Used	PL/SOP/06	Qualification Criteria	MDE-MP-SQC(01) , Rev : 04
Instrument Type	DSC-404C	Testing Temperature (°C)	RT to 1027°C(1300K)
Temperature(°C)	28.1, 28.3 ±1	Relative Humidity (%)	29, 29 ±2
Test Type	Sample ID	Test Results	
Specific Heat Capacity (Cp)	RD-19C/Cp/01	1.4765 J/gK at 427.0°C	
		2.4205 J/gK at 1027.0°C	
	RD-19C/Cp/02	1.6070 J/gK at 427.0°C	
		2.1803 J/gK at 1027.0°C	
	RD-19C/Cp/03	1.5966 J/gK at 427.0°C	
		1.7522 J/gK at 1027.0°C	
Test Status	Qualified for above mentioned Test i.e. ≥ 1230 J/Kg.K (1.230J/g.K) at 427.0°C (700K) & ≥ 1690 J/Kg.K (1.690J/g.K) at 1027.0°C (1300K)		
Remarks	i) These results apply to the items tested as received and mentioned above. ii) This test report shall not be reproduced except in full without the written approval of the laboratory. iii) The graphs of the samples analyzed are attached herewith. iv) Expanded Uncertainty is ± 0.101 J/gK at 300°C, ± 0.007 J/gK at 1000°C & ± 0.099 J/gK at 1400°C at 95% confidence level & k=2.		

Test Performed By:	Imran Shahzad ARO	13/05/2022	Contact No. 3301
Verified By:	Haseeb Ahmad Khan AM(T)		Contact No. 3301
Reviewed By:	Amir Khan HoD (CA)	13/05/22	Contact No. 3301
Approved By:	Engr. Dr Nabi Bakhsh PD (ME)	13.5.22	Contact No. 3141

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Prepared By: I/C/GL/SRO	Reviewed & Recommended By: GM/HoD	Approved By: PD/GMR
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